Vegetarian diets and blood pressure among white subjects: results from the Adventist Health Study-2 (AHS-2)

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Objective Previous work studying vegetarians has often found that they have lower blood pressure (BP). Reasons may include their lower BMI and higher intake levels of fruit and vegetables. Here we seek to extend this evidence in a geographically diverse population containing vegans, lacto-ovo vegetarians and omnivores. Design Data are analysed from a calibration sub-study of the Adventist Health Study-2 (AHS-2) cohort who attended clinics and provided validated FFQ. Criteria were established for vegan, lacto-ovo vegetarian, partial vegetarian and omnivorous dietary patterns. Setting Clinics were conducted at churches across the USA and Canada. Dietary data were gathered by mailed questionnaire. Subjects Five hundred white subjects representing the AHS-2 cohort. Results Covariate-adjusted regression analyses demonstrated that the vegan vegetarians had lower systolic and diastolic BP (mmHg) than omnivorous Adventists (β =–6·8, $P<0·05$ and β = –6·9, $P<0·001$). Findings for lacto-ovo vegetarians (β = –9·1, $P<0·001$ and β = –5·8, $P<0·001$) were similar. The vegetarians (mainly the vegans) were also less likely to be using antihypertensive medications. Defining hypertension as systolic BP > 139 mmHg or diastolic BP > 89 mmHg or use of antihypertensive medications, the odds ratio of hypertension compared with omnivores was 0·37 (95 % CI 0·19, 0·74), 0·57 (95 % CI 0·36, 0·92) and 0·92 (95 % CI 0·50, 1·70), respectively, for vegans, lacto-ovo vegetarians and partial vegetarians. Effects were reduced after adjustment for BMI. Conclusions We conclude from this relatively large study that vegetarians, especially vegans, with otherwise diverse characteristics but stable diets, do have lower systolic and diastolic BP and less hypertension than omnivores. This is only partly due to their lower body mass.